Source: `deliverables/PHASE12_STRONGEST_THEOREM.md` (repository dewarjoseph/-)

The strongest results of Phase XII (one page)

Setting.

- (G) $G = (V, E)$ is a countable, locally finite multigraph without loops, with every degree $\le d$.
- (U) Each oriented edge carries $U(v\to w) \in U(n)$, with $U(w\to v) = U(v\to w)^{-1}$.
- (H) $H_U = D_U\dagger D_U = \mathrm{Deg} - A_U$ on $\ell^2(V; \mathbb{C}^n)$, bounded, with domain all of ℓ^2, and $E_*(U) := \inf \sigma(H_U) = \inf$ over finitely supported $\psi \neq 0$ of $q_U(\psi)/\|\psi\|^2$.
- For Chair44: G is the panel multigraph of a chair tiling, $U(a\to b) = (\sigma\cdot M_a{}^{\mathsf T} e_k)(\sigma\cdot M_b{}^{\mathsf T} e_k) \in Q_8$, $\mathrm{Deg} \equiv 24$, and $E_*(r) = \inf$ over the hull of rule r.

Theorem 1 (support-sensitive two-sided principle; corrected from Phase XI)

(1a) $E_* = \inf\{\lambda_{\min}(P_K H_U P_K) : K$ a finite legal patch$\} = \lim_R D_R$, with $D_R \downarrow$.

- E_* is a function of the set of legal environments, and order-reversing in it.
- It does not depend on frequencies.
- Nothing beyond (G), (U), (H) is assumed. Neither (G) nor property A gives a rate.

(1b) For every partition of unity $\Sigma_j \chi_j^2 = 1$ with finite pieces K_j and $c_e = \Sigma \chi_j(v)\chi_j(w) > 0$, the identity $q_U = \Sigma_j \langle \chi_j\psi, (\mathrm{Deg} - \tilde A_U)\chi_j\psi\rangle$ holds exactly. Hence

$$\inf_j \lambda_{\min}(P_{K_j}(\mathrm{Deg} - \tilde A_U)P_{K_j}) \le E_* \le D_R,$$

and the lower side is $\ge \inf_j \lambda^D(K_j) - \varepsilon(\chi)$, with $\varepsilon(\chi) = \sup_v \Sigma_{e\ni v}(1/c_e - 1)$.

(1c) Rates.

- $\varepsilon \le 16 d C_D^2/R^2$ on uniformly doubling graphs, for $R \ge 4C_D$.
- $\varepsilon \le K_D(\rho) S_c/\rho^2$ for a proper realisation in $\mathbb{R}^D$. Here the pieces are Euclidean balls of radius $\rho > j_D\, h/\sqrt{(2D)}$, and $K_D(\rho) = (j_D^2/2D)/(1 - j_D^2 h^2/(2D\rho^2))$.
- R^{-2} is optimal ($\mathbb{Z}^D$). Property A alone gives no power law: the lamplighter $\mathbb{Z}_2 \wr \mathbb{Z}^k$, $k \ge 2$, has rate $\ge c_k R^{-2/k}$, so no exponent works for every property-A graph.

(1d) Boundary conditions and coverage.

- Neumann windows are not compressions. On the π-flux ladder they converge to $2 - \sqrt{2} < 3 - \sqrt{5} = E_*$.
- A lower bound computed over a strict subset of the legal environments can be false (ladder at $W = 8$: 0.49054 claimed, $\le$ 0.40791 true).

Theorem 2 (Orbit Lemma; new) [P, with every finite step checked exhaustively]

For a frame rule d with child factors h_c and Q in the chair's point group S_3:

- let $d\cdot Q$ have factors h_{cQ};
- let $\rho_Q(M, T) = (MQ, T)$;
- let conj_Q be the S_3-conjugation of rules.

Then:

- **(a)** $\sigma_{dQ} = \rho_Q \circ \sigma_d$, and ρ_Q preserves every tile's cube set.
- **(b)** ρ_Q is a global gauge transformation of every patch: $U^{\rho_Q} = W_Q\, U\, W_Q\dagger$, with $W_Q = x/|x|$ for an explicit integer quaternion x.
- **(c)** $T_1(dQ) \cong T_1(d)$ (unitarily), and the shadows are permuted, $\chi_k \mapsto \chi_{Q(k)}$.
- **(d)** If $\mathrm{conj}_Q(d) = d$, then $T_L(dQ) \cong T_L(d)$ for all L, and **$E_*(dQ) = E_*(d)$**.
- **(e)** Otherwise $\sigma_d \rho_Q \neq \rho_Q \sigma_d$, so the argument stops at level 1, and the equivalence can fail from level 2 on (certified for 1387 against 539288, corollary 3). It need not fail: chains of moves (d) and conjugations relate some such pairs (corollary 5).

The same holds verbatim for the cube family with the θ-connections, for proper Q and all θ. Improper Q map θ to −θ, so for them it holds where U_{−θ} ≅ U_θ: at the Pauli point θ = π/2, but not at θ = π/5 (level-1 bottoms split by parity; other θ not determined).

Certified corollaries [C]

1. **E_*(1387) ≥ 9.1104981** (radius 20, 24 571 legal balls; reproduced by an implementation sharing no code with the certifying one). E_*(1387) ∈ [9.1104981, 9.3616608] (the upper end on the pinned level-7 supertile, 2 097 152 tiles).
2. **The maximiser of E_* over all 6^8 = 1 679 616 Chair44 frame rules is one of 1387, 539311, 539291.** Every other rule has an exact rational upper bound below a certified lower bound: of 1387 (radius 20), or, for 1389, of 539311 (the preregistered radius-24 certificate, 39 117 legal balls: E_*(539311) ≥ 9.2083494 > 9.2083453 ≥ E_*(1389)). E_*(539311) ∈ [9.2083494, 9.3817818].
3. **The six-member S_3 orbit at the top of the frontier is not a symmetry of the operator.** E_*(539288) ≤ 9.0982535 < 9.1104981 ≤ E_*(1387), with 539288 = 4753·(0 2 1), although T_1 of the two rules are unitarily equivalent.
   - So **no function of level-one data determines E_*** (Theorem 3.1 of the report).
4. E_*(539315) = E_*(FR30): 1386 is fixed by conj_{(x y)}, an instance of (d).
5. **The lemma's moves are complete at level 3.** Inside right orbits, the exact level-3 spectral coincidences are precisely the closure of conjugation and the stabiliser clause. There are 1612 tied pairs among 1 401 660, with 0 unexplained and 0 related-but-split. Beyond level 3 this completeness is a conjecture.

What this does not establish

- **No global optimum.** 539311 now has the highest certified lower bound and the evidence points to it, but 1387 and 539291 are not excluded:
  - the frozen level-7 prediction held (Y_7(539311) − Y_7(1387) = 0.020121, order unchanged) [E];
  - two frozen extrapolations agree that E_*(539311) > E_*(1387) by about 0.012–0.020 [X].
  - No certificate separates them: IMS would need radius 67–91.
- **No monotonicity of single-child interventions** [E]. Around FR30 they raise Λ_6 for 2 of 22 neighbours and lower it for 20; the language-inclusion comparison (1a) applies to none of the 60 non-trivial neighbours.
- **No quantitative law for the splitting** inside an orbit. The conjugation defect was preregistered as such a law and **failed** (Spearman −0.06).
- **No sharp curvature bound.** The support-sensitive transport bound is valid (1.69926 for the orbit of 1387) but captures under a fifth of the floor. Lengths, overlaps, eigenangles and plaquette classes cannot be sharp: the $\mathbb{Z}^3$ pair has identical inputs and floors separated by 0.18165.
- **No proof-assistant formalization.** Lean was unavailable; the finite core is checked by exact computation.
- **No physical law, observable or particle spectrum.**